



USA TRIATHLON EDUCATION

7-Zone Training System for Running

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Overview

The 7-zone system anchors every running zone to LT1 and LT2, so each boundary lands on a meaningful physiological point instead of inside a zone. Zones 1 and 2 sit at or below LT1, Zones 3 and 4 sit between LT1 and LT2, and Zones 5 to 7 sit above LT2.

In running, LT2 is threshold pace. Running zones need their own thresholds and their own testing. Do not carry zones over from the bike or the pool, because running power and cycling power are not on the same scale and running heart rate behaves differently.

The seven running zones

Zone	Pace definition	Position	Purpose
1	25% to 10% slower than LT1 pace	Below LT1	Recovery and easy movement
2	10% slower than LT1 pace up to LT1 pace	Up to LT1	Aerobic base: capillarization, blood volume, fat use, connective tissue
3	LT1 pace to the midpoint of LT1 and LT2 pace	Between LT1 and LT2	Tempo, moderate aerobic work
4	Midpoint of LT1 and LT2 pace up to LT2 pace	Just below LT2	Sub-threshold, building toward threshold
5	100% to 93% of LT2 pace	Just above LT2	Above-threshold work
6	93% to 85% of LT2 pace	Higher intensity	Anaerobic capacity, lactate tolerance
7	85% to 72% of LT2 pace	Near VO2 and maximal	Maximal efforts and sprints

Zones 5 to 7 are fixed multiples of LT2 pace, so they are the same for every athlete. Zones 1 to 4 shift with the athlete's LT1 percentage: about 70% of LT2 for Age-Group, 75% for Performance, and 80% for Elite. Zones 3 and 4 split the gap between LT1 and LT2 at the midpoint of the two paces, not the midpoint of the speeds.

Percent of LT2 speed (higher is faster):

Zone	Age-Group (LT1 = 70%)	Performance (LT1 = 75%)	Elite (LT1 = 80%)
1	56 - 64%	60 - 68%	64 - 73%
2	64 - 70%	68 - 75%	73 - 80%
3	70 - 82%	75 - 86%	80 - 89%
4	82 - 100%	86 - 100%	89 - 100%
5	100 - 107.5%	same	same
6	107.5 - 117.6%	same	same
7	117.6 - 138.9%	same	same

Percent of LT2 pace (a bigger number is slower):

Zone	Age-Group (LT1 = 70%)	Performance (LT1 = 75%)	Elite (LT1 = 80%)
1	179 - 157%	167 - 147%	156 - 138%
2	157 - 143%	147 - 133%	138 - 125%
3	143 - 121%	133 - 117%	125 - 113%
4	121 - 100%	117 - 100%	113 - 100%
5	100 - 93%	same	same
6	93 - 85%	same	same
7	85 - 72%	same	same

Check whether the system you enter these into wants percent of threshold speed or percent of threshold pace. The two tables are not interchangeable. For heart rate, Zones 1 to 4 follow the same structure, with LT1 heart rate at 83, 86, or 89% of threshold heart rate depending on category. Zones 5 to 7 are built from the gap between threshold heart rate and maximum heart rate rather than as a percentage of threshold heart rate.

Zone 2 and the slowest sustainable pace

Many athletes cannot run at true Zone 2. Even their slowest running pace pushes heart rate above the target, because running spikes heart rate more than swimming or cycling. That is common and acceptable. Your options:

- **Run at the slowest sustainable pace at which running biomechanics are not negatively affected.** Slower is not better once form breaks down. Find the pace where stride, posture, and arm carriage still look right.
- **Use a walk-run protocol**, for example 3 minutes running and 3 minutes walking, so the average intensity stays in Zone 2.
- **Speed walk** when running is too intense.
- **Let swimming and cycling carry the base.** Zone 2 emphasis matters most early in the training cycle, and pushing higher intensity then prevents proper base development.

Work-to-rest ratios for running

The harder the interval, the longer the rest it needs. Insufficient rest also degrades biomechanics, so a breakdown in running form during a session tells you the rest was too short.

Zones	Energy system	Work-to-rest guidance for running
1-2	Aerobic	Minimal byproducts and quick recovery. Continuous running or walk-run, with moderate recovery between sessions
3-4	Aerobic into anaerobic	Rest lengthens as pace approaches LT2. Manage session density and weekly volume
5-6	Anaerobic	Substantially more rest so lactate can partially clear. For example, the 4 x 1600 m test uses 90 seconds of recovery at 90 to 100% effort (make sure athletes have been signed off by their doctor for physical exercise, and only complete this test once the athlete is well conditioned to training)
7	Alactic / ATP-CP	Sprints under 10 seconds need near-full recovery, around 1:5 to 1:8 work to rest

Hill repeats for strength endurance. In a 5 x 1 km hill repeat session, allow generous recovery on the descent, for example 6 to 7 minutes of descent for a climb of 3.5 to 4 minutes. Strength endurance is muscular, so the legs do not fully recover on the way down regardless of pace.

Warm up well for Zone 4 and above. A 25 to 30 minute warm-up makes a substantial difference to session quality and injury risk: an easy jog, dynamic stretching, short mechanics efforts, then shuttle runs to prime the switch between aerobic and anaerobic output. If time is short, protect the warm-up before the warm-down.

Coaching notes for running

- **Retest often enough.** Re-establish run thresholds every 3 to 6 months and reset zones after each test, in the first week after an adaptation week.
- **Watch mechanics at speed.** Technique breakdown tends to show up at higher zones. Video analysis during progressively faster intervals is worth the time.
- **Use running power as a personal signal.** It is estimated rather than measured, so use it to hold effort steady and compare an athlete to their own past data, not to compare athletes or disciplines.
- **Match zones to the athlete.** Long-course runners may rarely use Zones 5 to 7, but keeping them in the system makes it easy to adapt if the athlete wants to race shorter.